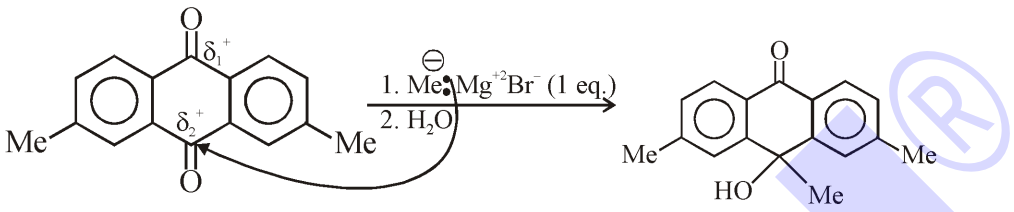
SECTION–I : (i) Only One option correct Type

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

3(–1)

1. (B)

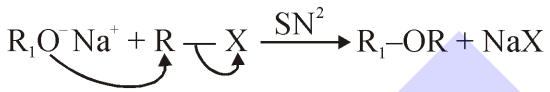
Sol.

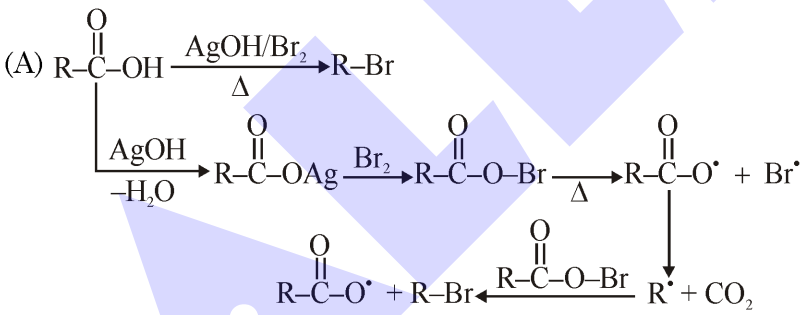


$\delta_2^+ > \delta_1^+$ because of +H effect of Me group applicable on o/p position $\delta_1^+\downarrow$

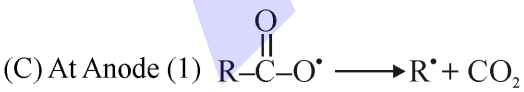
2. (C)

Sol. Williamson's ether synthesis is SN^2 reaction, do not involve free radial formation.



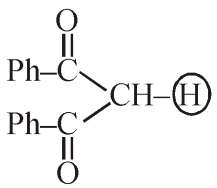
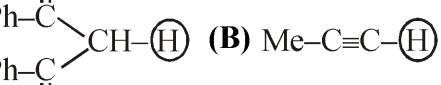
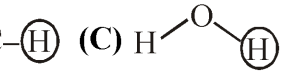


(B) $R^\bullet + R^\bullet \rightarrow R\text{–}R$



(2) $R^\bullet + R^\bullet \rightarrow R\text{–}R$

3. (D)

Sol. **(A)**  **(B)**  **(C)** 

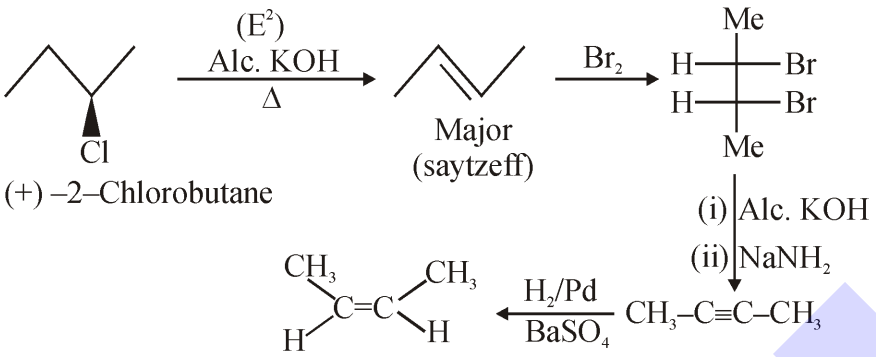
Compound (A), (B) and (C) have H-atoms acidic enough to show acid base reaction with MeMgBr but

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(D) has less acidic H-atom so no reaction with Grigneard reagent.

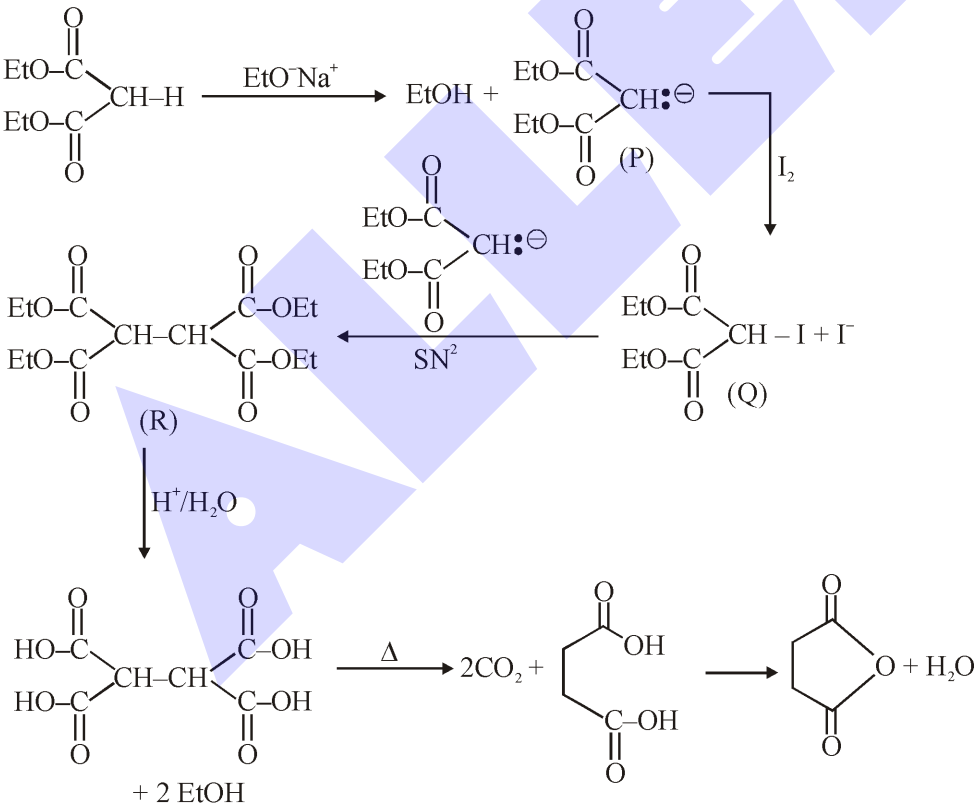
4. (C)

Sol.



5. (B)

Sol.



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6. (C)

Sol. $CH_3-CH_3 \xrightarrow[h\nu]{Cl_2} CH_3-\underset{Cl}{CH_2} \xrightarrow[SN^2]{aq.KOH} CH_3-\underset{OH}{CH_2}$

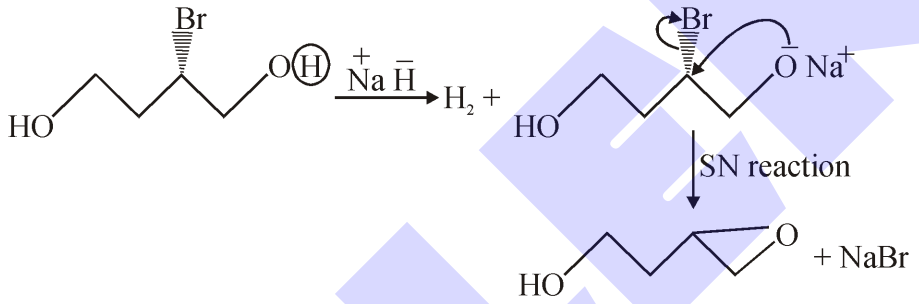
$CH_3-\underset{OH}{CH_2} \xrightarrow{(Oxidation)\ KMnO_4/NaOH} CH_3-\underset{O}{\overset{}{C}}-OH$

$CH_3-\underset{O}{\overset{}{C}}-OH \xrightarrow[CaO/\Delta]{NaOH} CH_4$

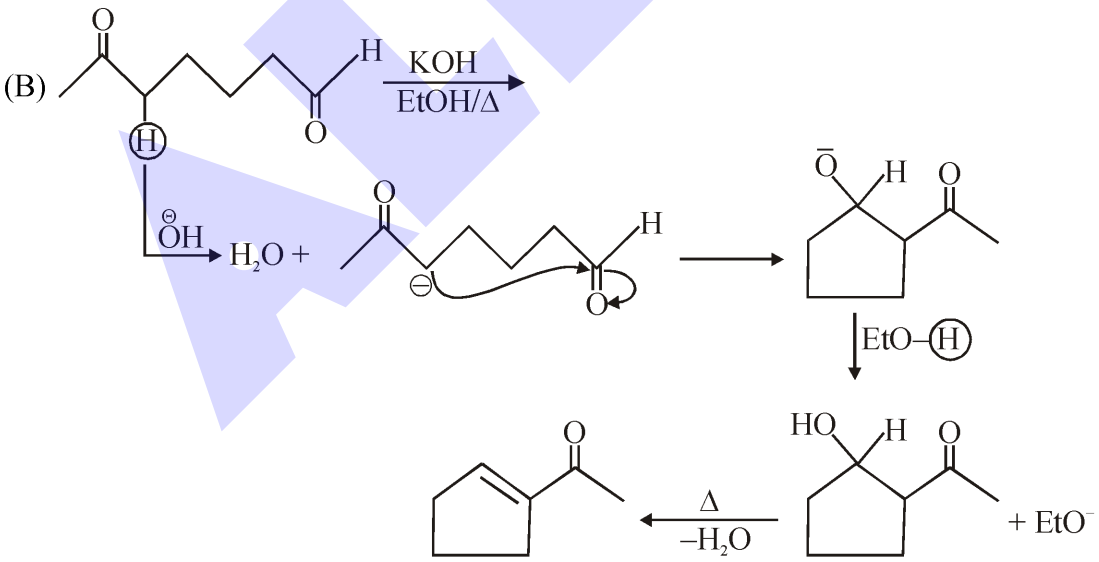
Sodalime decarboxylation

7. (A,B,C,D)

Sol. (A)

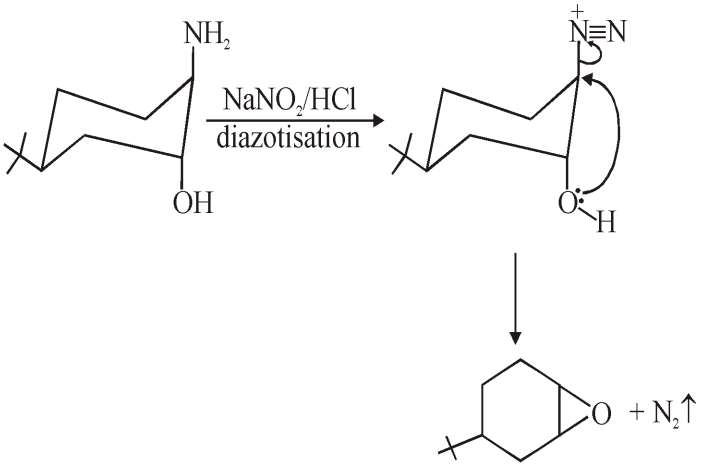


(B)

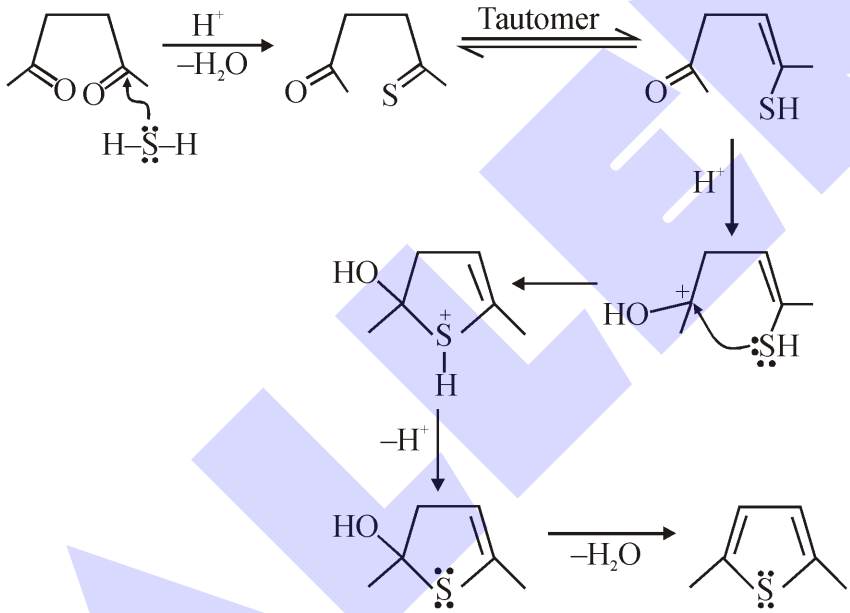


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(C)



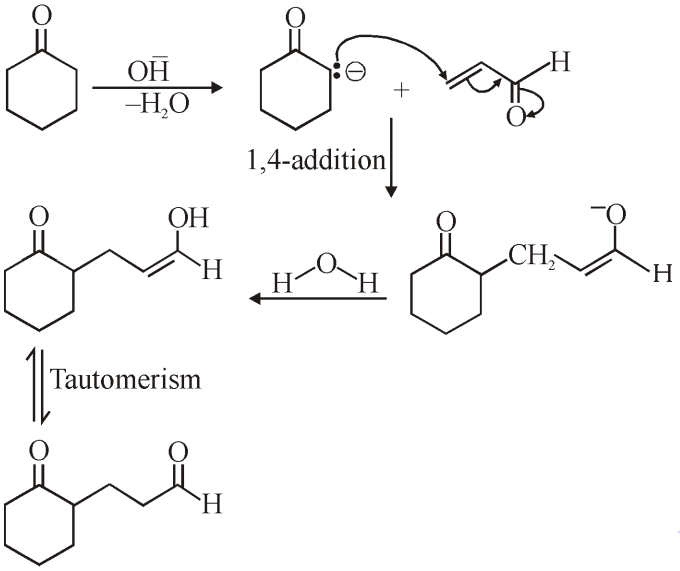
(D)



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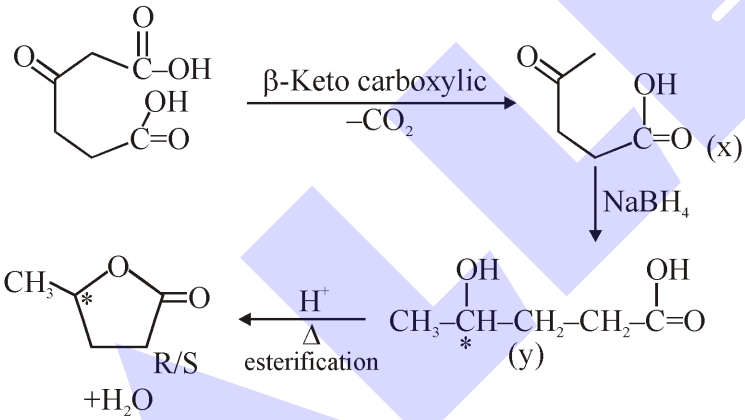
8. (BC)

Sol.

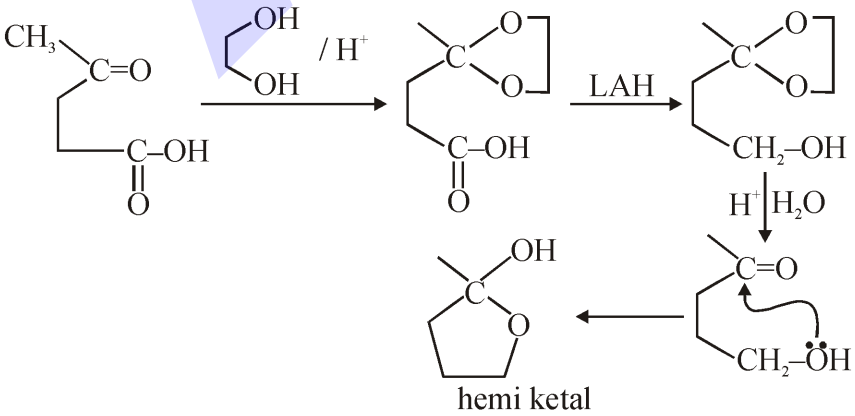


9. (A,B,C,D)

Sol.



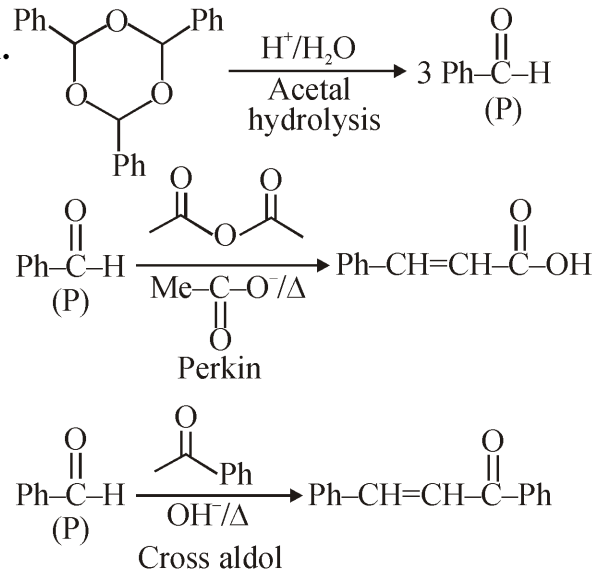
Esters are sweet smelling compounds.



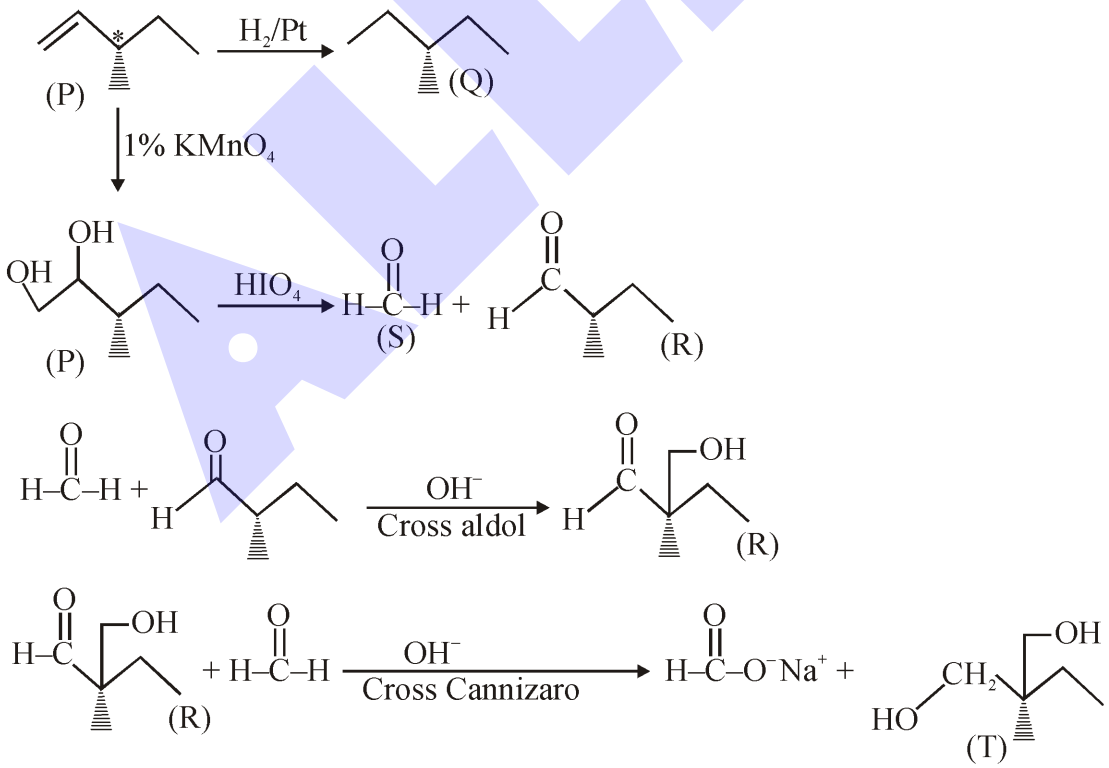
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10. (A,B)

Sol.



Paragraph Q. NO. 11 & 12

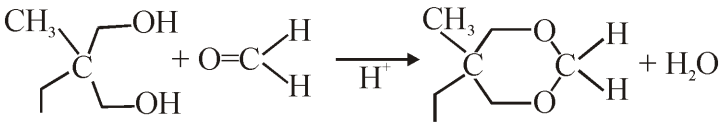


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11. **(A,C,D)**

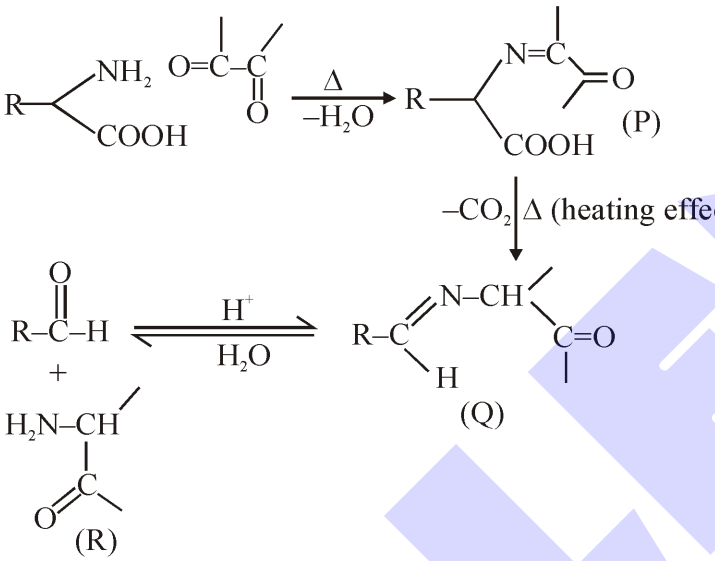
12. **(C)**

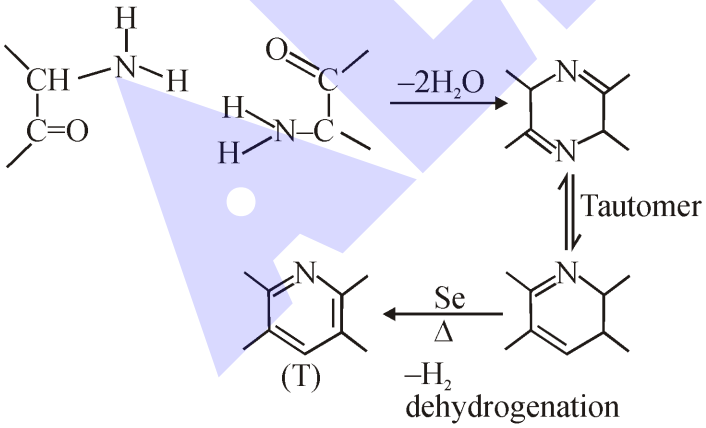
Sol.



13. **(A)**

Sol.





14. **(D)**

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Matching list Type

15. **(P)** G.I ✓ (Q) G.I ✓

Tautomerism ✓ Tautomerism × (0 α-H)

Optical × Optical ✓

(R) G.I ✓ (Q) G.I ×

Tautomerism ✓ Tautomerism ✓

Optical ✓ Optical✓

Section -III

(1) Silver mirror test **(Ans.6)**

(i) ✓ (ii) ✓ (v) ✓ (vii) ✓ (Hemi acetal) (ix) ✓ (x) ✓

(2) $R{-}O{-}H + CH_3{-}\overset{\overset{O}{\|}}{C}{-}Cl \xrightarrow{-HCl} R{-}O{-}\overset{\overset{O}{\|}}{C}{-}CH_3$ Increase in mass = 42 = 43–1 **(Ans.6)**

$R{-}N\begin{matrix}H\\H\end{matrix} + CH_3{-}\overset{\overset{O}{\|}}{C}{-}Cl \xrightarrow{-HCl} R{-}N(\overset{\overset{O}{\|}}{C}{-}CH_3){-}H$

No. of –OH + number of $-NH_2$ (in a molecule) = 3

(i) ✓ (ii) ✓ (vi) ✓ (viii) ✓ (ix) ✓ (x) ✓

(3) V – L – G – A (Total = 4 structure possible)

V – G – L – A

L– G – V – A

L – V – G – A

(4) Cellose + $nH_2O \longrightarrow$ Glucose
Segment

$\underbrace{R_1{-}O{-}R_2}$ + $H_2O \longrightarrow R_1{-}OH + HO{-}R_2$

Increase in mass per glycosidic linkage = 18

Total glycoside linkage = x

x × 18 = 1620 – 1476

x = 8

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